# Core-shell nanostructures involving MoS2

"Does it possible to have MoS2 as core and SiO2 as a shell ? or other structure with this sort?"
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Yes — **a $MoS_2$ core with a $SiO_2$ shell is structurally plausible, and the retrieved evidence also shows the reverse architecture ($SiO_2$ core with $MoS_2$ shell) and even ternary stacked structures that include both materials.** [11] [69] [70] [71] [72]

## What the evidence shows

| Structure type | What was reported | Why it matters |
|---|---|---|
| **$SiO_2$ core / $MoS_2$ shell** | **$SiO_2$@$MoS_2$ nanospheres** were prepared by hydrothermal reaction, with **$MoS_2$ nanosheets growing on the outside surface of $SiO_2$** [73] | Confirms that a core-shell arrangement between these two materials is feasible, with $MoS_2$ as the outer active layer [73] |
| **$MoS_2$ core / $SiO_2$ shell** | **Silica-coated $MoS_2$ nanocomposites with hierarchical and core-shell structure** were synthesized, where **$SiO_2$ grew uniformly on $MoS_2$ nanoflowers** [70] | Directly supports the exact arrangement you asked about: $MoS_2$ as the core and $SiO_2$ as the shell [70] |
| **$MoS_2$-encapsulated $SiO_2$** | A TENG study reported **$MoS_2$-encapsulated $SiO_2$ nanoparticles** with a **thin $MoS_2$ layer uniformly grown on $SiO_2$** [11] | Shows the same materials can be combined in the opposite direction for triboelectric devices, especially for humidity resistance [11] |
| **Ternary layered structure** | **$H_2Ti_2O_5$@$MoS_2$@$SiO_2$** core-shell nanoparticles were made, where **$SiO_2$ acted as an insulating cladding layer** and $MoS_2$ supplied conductivity [71] [72] | Shows that $MoS_2$ and $SiO_2$ can be combined in multi-shell architectures, not just simple binary shells [71] [72] |

## For TENG use

For TENGs, the most relevant design logic is that **$MoS_2$ provides charge-trapping/triboelectric activity**, while **$SiO_2$ can act as an insulating shell that suppresses excessive conductivity and helps retain charge**. That pattern is already reflected in the PVDF composite literature and in the humidity-resistant TENG example [11] [69].

So the short answer is: **yes, $MoS_2$-core/$SiO_2$-shell is possible, and it is consistent with reported core-shell synthesis strategies; in TENGs, it would be a reasonable architecture if the goal is to balance charge trapping with leakage suppression**. [11] [69] [70]

**Bottom line:** the literature supports **both directions of $MoS_2/SiO_2$ core-shell design**, and for TENGs the most promising role of a **$SiO_2$ shell on $MoS_2$** is charge confinement and stability rather than making the shell electrically active. [11] [69] [70]

Confidence level: **High** — direct core-shell examples exist for both $MoS_2@SiO_2$ and $SiO_2@MoS_2$, including a TENG-relevant $MoS_2$-encapsulated $SiO_2$ system and a silica-coated $MoS_2$ system.

## References


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